

CNN-Adaptive Sliding Mode Control for Autonomous Differential-Drive Robots

Under Environmental Uncertainty — A Digital Twin Approach

Likhita Yerra Mohamed Oussama Bouriga Ahmed Ben Aissa Abdellahi El
Moustapha Thibault Goutorbe

PGE5 — AI Clinic | Supervisor: Prof. Vishvjit Thakar



31 August 2026

Environment → AI Agent → SMC Control → Robot

Outline

- 1 Introduction
- 2 Methodology
- 3 Results
- 4 Digital Twin Demo
- 5 Conclusion

Problem Statement

- Autonomous mobile robots must track trajectories under **sensor noise**, **external disturbances**, and **wheel slip**.
- Sliding Mode Control (SMC) is robust, but **fixed gains** cannot simultaneously optimise:
 - chattering suppression,
 - tracking accuracy,
 - disturbance rejection.
- Noisy conditions $\Rightarrow$ smoother control; disturbances/slip $\Rightarrow$ aggressive correction.

Research Question

Can a learning-based perception and control pipeline improve SMC adaptability for a differential-drive robot under heterogeneous uncertainty?

Three Controllers Compared

Classical SMC

- Fixed gains
- Baseline
- Struggles under combined uncertainty

CNN-Adaptive SMC

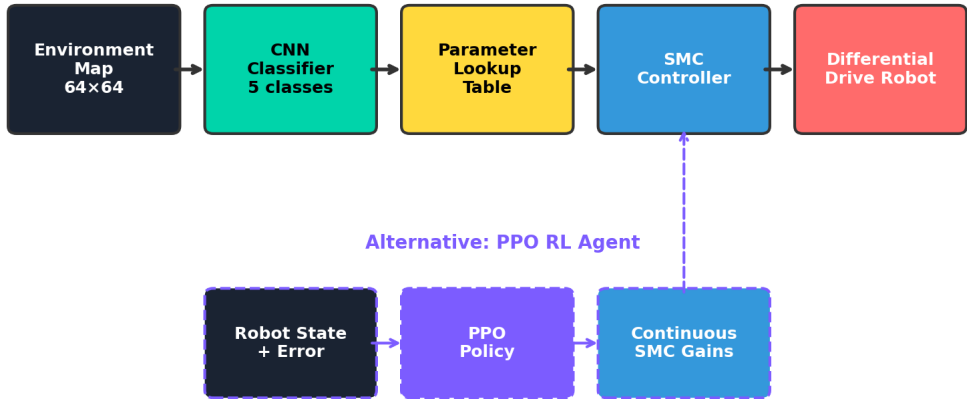
- 64×64 map $\rightarrow$ CNN
- Scenario $\rightarrow$ preset lookup
- Interpretable adaptation

RL Agent (PPO)

- Continuous policy
- Reward-driven presets
- Extension / baseline

System Architecture

System Architecture — CNN-Adaptive SMC Pipeline



Differential-drive unicycle

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \omega$$

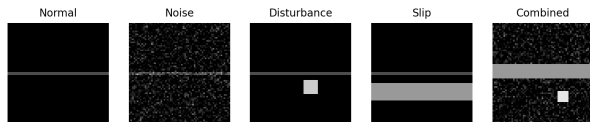
- $\Delta t = 0.01 \text{ s}$, $L = 0.3 \text{ m}$
- Reference speed $v_r = 0.3 \text{ m/s}$, $T = 20 \text{ s}$
- Trajectories: straight, circle, S-curve

Five uncertainty scenarios

Scenario	Perturbation
Normal	None
Noise	$\sigma_{xy} = 0.02 \text{ m}$ on pose
Disturbance	Push $+0.4 \text{ m}$ in y at $t = 8 \text{ s}$
Slip	30% velocity loss $t \in [10, 14] \text{ s}$
Combined	All three together

CNN Environment Classifier

CNN Environment Map Samples (64×64 grayscale)



EnvironmentCNN ($\approx 534k$ params)

- Input: $1 \times 64 \times 64$ grayscale map
- Output: 5 scenario classes
- 3 conv blocks + FC + dropout
- **100%** test acc. (simple maps)
- **95.1%** on realistic maps
- Pre-mission classification → SMC preset

CNN-adaptive SMC

- Map class $\hat{c} \mapsto$ preset $\pi_{\hat{c}}$
- Exponential filter:
$$\pi_k = (1 - \beta)\pi_{k-1} + \beta\pi_{\hat{c}}$$
- Hand-tuned per scenario $(\lambda, k, \varphi, \alpha_\omega)$
- Lyapunov UUB analysis grounds design

PPO RL baseline

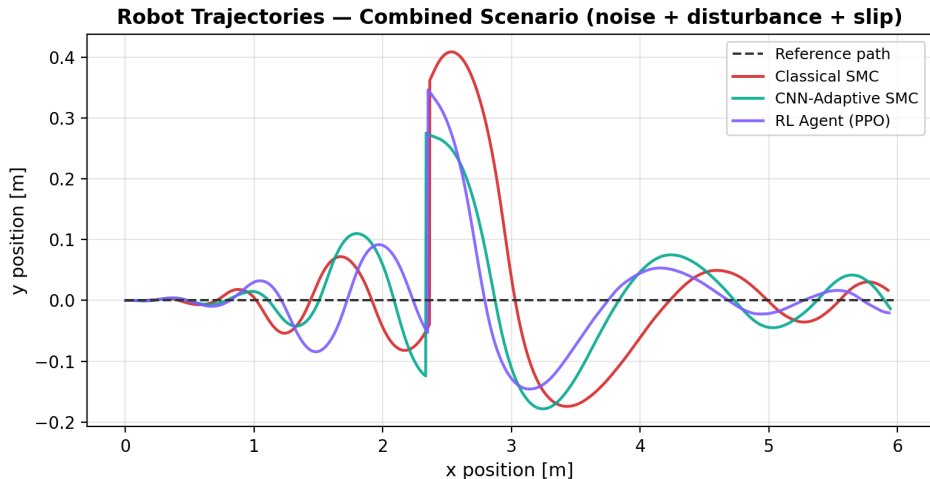
- Gymnasium env + domain randomisation
- Obs.: error, error rate, chattering, flags
- Action: select preset every 0.2 s
- Reward: $r = -e - 0.05j_\omega - 0.001|\omega_c|$
- 80k training steps

Headline Results

Metric	Improvement	Context
Chattering index	35.7% reduction	Sensor noise
Final tracking error	14.4% improvement	After disturbance
Final tracking error	32.7% improvement	Wheel slip
CNN vs. oracle	17.9 mm vs. 15.6 mm	Near-oracle recovery
Realistic-map CNN	95.1% accuracy	Deployability

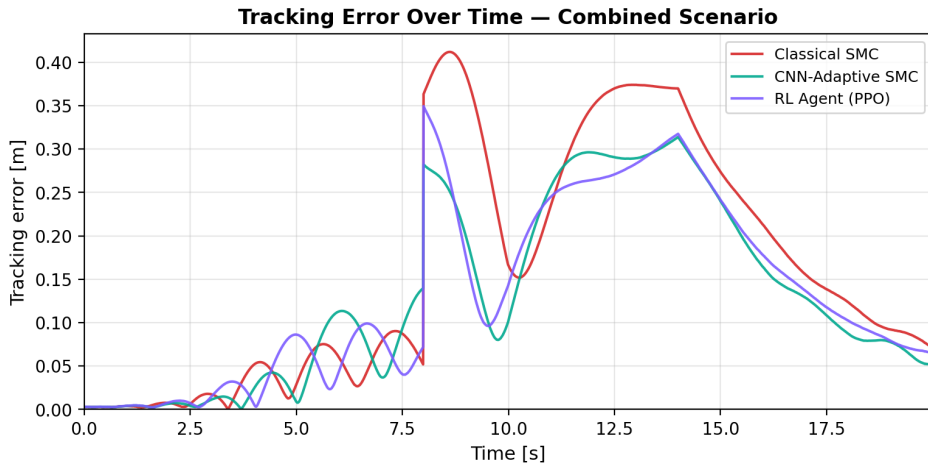
Trade-off: noise scenario widens steady-state error (smoothness vs. accuracy per Lyapunov bound).

Trajectory Comparison — Combined Scenario



Impulsive push at $x \approx 2.4$ m; adaptive controllers recover more tightly to the reference.

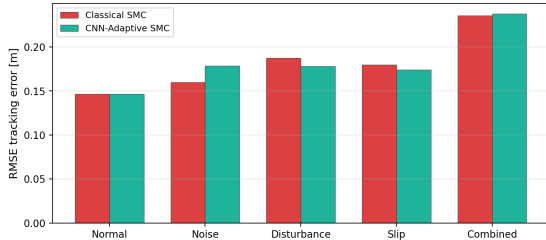
Tracking Error Over Time



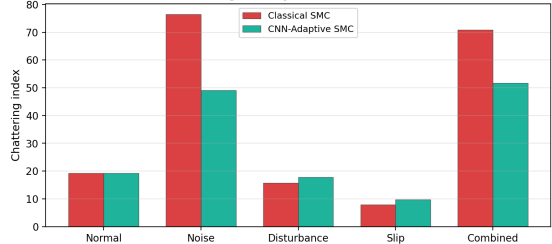
Full 20 s episode: disturbance at $t = 8$ s, slip window $t \in [10, 14]$ s.

Quantitative Comparison

RMSE Tracking Error per Scenario (Table 3)

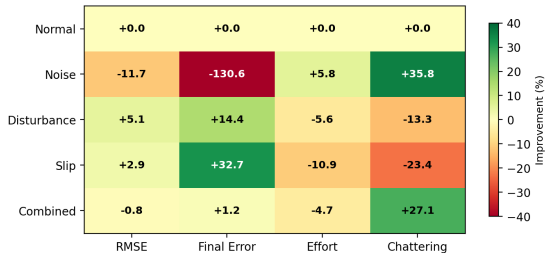


Chattering Index per Scenario (Table 3)

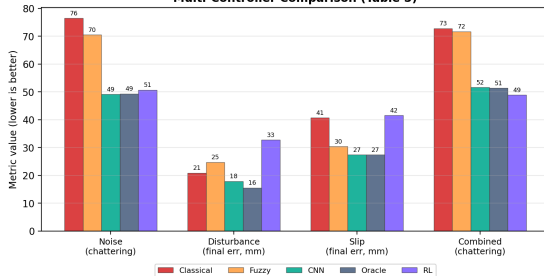


Improvement Heatmap & Multi-Controller Benchmark

CNN-Adaptive Improvement over Classical SMC (%)

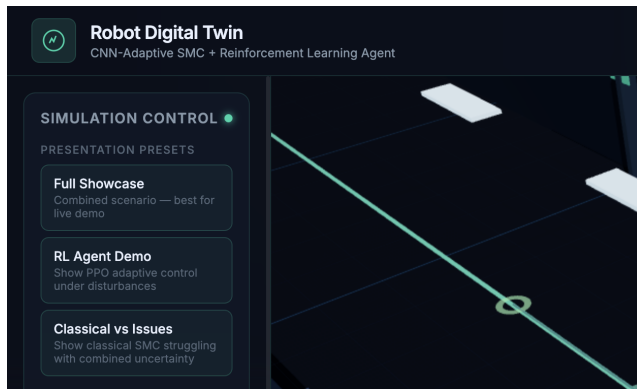


Multi-Controller Comparison (Table 5)



CNN-adaptive and oracle presets achieve largest chattering reductions; RL competitive under combined uncertainty.

Live Demonstration



Web-based 3D twin

- React + Three.js frontend
- FastAPI WebSocket backend
- Hospital corridor environment

Demo flow

- 1 `python run_digital_twin.py`
- 2 Open `localhost:8000`
- 3 Press **T** — Controller Tour
- 4 Compare tab — benchmark

Space = run/pause 1/2/3 = controllers

Conclusions

- Complete pipeline: synthetic maps $\rightarrow$ CNN $\rightarrow$ adaptive SMC $\rightarrow$ simulation $\rightarrow$ digital twin.
- CNN-adaptive SMC **reduces chattering** and **improves recovery** with interpretable presets.
- Near-oracle performance vs. fuzzy and RL baselines on published benchmark.
- Lyapunov UUB analysis provides theoretical grounding for scenario-specific φ .
- Real-time 3D twin enables live defense demonstration.

Future Work & Acknowledgement

Future work

- Real LiDAR occupancy grids
- Continuous on-line scenario inference
- Extended RL training + reward shaping
- Monte-Carlo evaluation + physical robot

Resources

- Code: github.com/LikhitaYerra/smc_cnn
- Report: AI Clinic submission
- Conference: DELCON 2026

We thank **aivancity** and Prof. Vishvjit Thakar for supervision and support.

Thank You

Questions?



PGE5 — AI Clinic | aivancity, Paris